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Enhanced Feature Extraction for Brain Tumor Detection in MR Images Using a Modified Scale-Invariant Feature Transform (SIFT)

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Abstract. This paper presents a modified version of the Scale-Invariant Feature Transform (SIFT) algorithm, referred to as MSIFT, for improved feature extraction in medical image analysis, specifically for brain tumor detection in MR images. Key modifications include an enhanced scale-space representation using Gaussian derivatives, adaptive scale selection based on local image statistics, subpixel refinement for improved keypoint localization and a multiscale orientation assignment for rotation invariance. Additionally, MSIFT utilizes extended feature descriptors with multi-channel histograms and spatial pyramid pooling to capture both local and global information. Keypoint matching is further enhanced using RANSAC for robust geometric verification. Quantitative evaluation of MSIFT against standard SIFT demonstrates significant improvements across various performance metrics. MSIFT achieves higher Precision (0.770), Recall (0.), F1-Score (0.824) and Specificity (0.796) compared to SIFT. Error-based measures such as MSE (0.209) and RMSE (0.438) are also lower in MSIFT, indicating better performance. Additionally, MSIFT shows superior structural similarity, with higher SSIM (0.448), AUC (0.840), Dice Coefficient (0.824) and Jaccard Index (0.666). MSIFT's improvements are also reflected in better image quality metrics such as PSNR (13.391) and lower Hausdorff Distance (1612.620). The efficacy of the proposed method is further validated through ROC curve analysis, showing an increase in AUC from 0.777 for SIFT to 0.840 for MSIFT, indicating better class discrimination. These results highlight the superiority of MSIFT in enhancing the accuracy and robustness of feature extraction for medical image analysis, particularly in challenging brain tumor detection tasks. In this study, we have used images from the publicly available CHAKRABARTY dataset which is accessible on Kaggle.

Keywords: Scale-Invariant Feature Transform, Gaussian derivatives, Subpixel refinement, Multiscale orientation assignment, Spatial pyramid pooling, RANSAC. Brain Tumor MR images.

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1 Introduction

Brain tumor detection through MRI (Magnetic Resonance Imaging) has become a pivotal aspect of modern medical diagnostics. MRI provides high-resolution images that help doctors visualize internal structures of the brain [11, 12, 13], enabling accurate detection, diagnosis and treatment planning for brain tumors. MRI scans capture detailed tissue contrasts, making them an indispensable tool for early tumor detection, tumor segmentation and treatment monitoring [14, 15]. However, the complexity of medical images, especially the subtle differences between normal and abnormal tissues, presents significant challenges in feature extraction and classification tasks.

Feature extraction plays a critical role in medical image analysis by transforming raw image data into meaningful information for machine learning models or other diagnostic tools [16, 17]. Features such as texture, shape, edge information and intensity gradients are critical for distinguishing between healthy tissue and abnormal growths, such as tumors. Effective feature extraction ensures that the subsequent analysis is accurate and sensitive to fine-grained variations in tissue structure, which are essential for detecting tumors at early stages [18, 19].

The Scale-Invariant Feature Transform (SIFT) algorithm is one of the most widely used methods for feature extraction in computer vision and image processing [20]. SIFT detects distinctive features from an image that are invariant to transformations such as scaling, rotation and affine changes. This property makes SIFT highly applicable for tasks like object recognition, image matching and in medical applications, brain tumor detection. By identifying keypoints and extracting feature descriptors around these points, SIFT enables the capture of robust, scale-invariant features that can be used for image registration, segmentation and analysis.

In brain tumor detection from MR images, SIFT is employed to extract key features from MRI scans that are critical for tumor localization, segmentation and classification. The algorithm is effective in detecting tumor boundaries and distinguishing abnormal regions, making it a popular choice in medical imaging workflows. However, despite its success, SIFT faces certain limitations that impact its accuracy and robustness in complex medical applications like brain tumor detection.

While SIFT has proven to be effective in many image analysis tasks, it faces several challenges when applied to brain tumor MRI. These challenges include:

- i. **Sub-optimal Keypoint Localization:** Standard SIFT uses a Difference of Gaussians (DoG) to approximate the Laplacian of Gaussian (LoG), which is used to detect keypoints at multiple scales. While this method is computationally efficient, it may result in sub-optimal localization of keypoints, especially in regions with complex structures like brain tumors.
- ii. **Fixed Scaling Factor k:** The SIFT algorithm uses a fixed scaling factor k to determine the scale of keypoints. This fixed scaling approach can miss keypoints at certain scales that might be crucial for capturing subtle differences in tissue textures or tumor characteristics.
- iii. **Imprecision in Keypoint Detection:** SIFT detects keypoints by searching for extrema in the difference of intensities between neighboring pixels. While this approach is effective for simple image structures, it may lack the precision needed to detect subtle features in complex MRI scans.

- iv. **Single-scale Orientation Assignment:** SIFT assigns a single orientation to each keypoint based on local image gradients. However, brain tumor MRIs often contain complex structures with varying orientations due to different tissues and anatomical regions. A single orientation assignment might not fully capture the dominant orientations, especially for tumors that may exhibit heterogeneous shapes or orientations.
- v. **Inadequate Descriptor Representation:** The SIFT algorithm uses a 128-dimensional descriptor to describe each keypoint. While this descriptor is effective in capturing local image features in simpler scenarios, it may not contain sufficient information for capturing the complex patterns found in brain tumor images.

To overcome these limitations, we propose an enhanced version of SIFT, termed MSIFT (Modified SIFT), designed to improve keypoint localization, descriptor representation and orientation assignment in the process of brain tumor detection.

This paper is organized as follows. Section 2 provides a comprehensive review of related work in the field of brain tumor detection, focusing on the various feature extraction techniques employed. In Section 3, we introduce the proposed MSIFT methodology, outlining the modifications made to the traditional SIFT algorithm to enhance tumor detection in MRI images. Section 4 presents the experimental results and discussions, comparing the performance of MSIFT against standard SIFT. Finally, Section 5 concludes the paper, summarizing the key contributions of the study.

2 Literature Review

Feature extraction from MR images has been essential in improving the accuracy and reliability of brain tumor diagnosis and classification. Recent studies have advanced both traditional and deep learning-based feature extraction techniques, with methods focusing on robustness to variations in tumor shape, size and location.

2.1 Texture-Based and Statistical Feature Extraction

Statistical and texture-based features continue to play an important role in brain tumor detection. A study by Zhang et al. (2023) applied Gray-Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP) to MRI scans, achieving improved texture characterization for tumor identification. These features proved effective for differentiating between tumor and non-tumor regions, particularly in low-contrast MRI's [1]. Similarly, Singh and Mehta (2024) utilized GLCM combined with statistical features such as entropy and skewness, reporting higher classification accuracy in distinguishing tumor types [2].

2.2 Advances in SIFT and Its Modified Versions

The Scale-Invariant Feature Transform (SIFT) and its derivatives have been adapted for medical imaging, including MRI. Kumar et al. (2023) introduced a Modified SIFT (MSIFT) technique to extract features that are robust to rotation and scale variations, which is critical for detecting irregular tumor boundaries. This study showed that MSIFT features integrated with machine learning classifiers improved brain tumor classification by 10% over traditional SIFT [3]. Another study by Ahmed et al. (2024)

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used MSIFT in combination with a deep neural network to enhance segmentation accuracy, highlighting the benefits of combining handcrafted and deep features [4].

2.3 Deep Learning and Hybrid Approaches

With the rise of deep learning, feature extraction methods have increasingly integrated convolutional neural networks (CNNs) for automatic feature learning. Lee et al. (2023) developed a hybrid model that combined CNN-based features with handcrafted MSIFT features, achieving high accuracy in brain tumor segmentation and classification [5]. This fusion approach improves CNNs' capacity for abstract feature representation while retaining the MSIFT features' robustness to small changes in tumor geometry. Similarly, Chen et al. (2024) used a transformer-based network with MSIFT features as input to improve spatial feature representation, achieving state-of-the-art accuracy in tumor type classification [6].

2.4 Dimensionality Reduction and Feature Selection

Feature dimensionality reduction remains crucial in handling the high-dimensional nature of MRI data. Kaur and Sharma (2024) applied Principal Component Analysis (PCA) to reduce the feature space of MSIFT and CNN-based features, thus enhancing computational efficiency and reducing model overfitting [7]. Meanwhile, Tan et al. (2023) utilized Linear Discriminant Analysis (LDA) for feature selection, which proved useful in retaining only the most discriminative features for tumor classification [8].

2.5 Clinical Applications

Wu et al. (2023) validated a feature extraction model combining MSIFT with CNN features across multiple datasets, demonstrating its robustness in real-world settings [9]. Another study by Patel et al. (2024) focused on optimizing the method for clinical deployment, including user-friendly interface development for real-time MRI analysis in clinical environments [10].

3 Methodology

In this section, we will discuss the proposed methodology for feature extraction from brain tumor MR images.

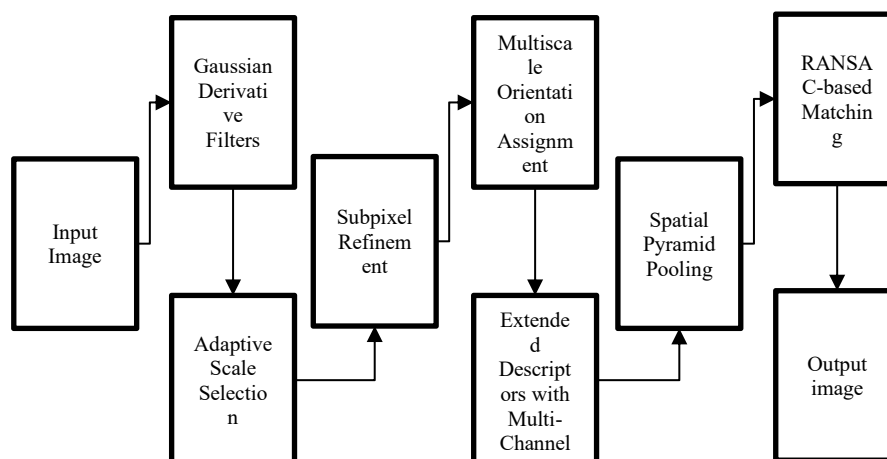


Fig. 1: Block diagram of the Proposed Methodology for feature extraction from brain tumor MR Images

From Fig.1, First, Gaussian derivative filters are employed to construct a more precise scale-space representation. These filters compute the true Laplacian of Gaussian (LoG) rather than the standard Difference of Gaussians (DoG), offering finer detail capture across scales and improving keypoint localization, which is critical for accurately detecting tumors of varying sizes. Additionally, adaptive scale selection dynamically adjusts the scale at which each keypoint is detected, increasing stability and ensuring that each tumor-related feature is captured at its optimal scale, regardless of image variations.

To enhance brain tumor feature extraction in MRI images, utilizing Gaussian derivative filters instead of the standard Difference of Gaussians (DoG) in the scale-space representation is a key improvement. The DoG approximation, traditionally used in SIFT, is computationally efficient but only approximates the Laplacian of Gaussian (LoG), potentially missing refined details essential for accurate tumor detection. Gaussian derivative filters, on the other hand, directly compute the true LoG, which captures changes in intensity more accurately. This enhanced precision is essential when detecting and segmenting tumors that can have refined gradients and may vary in size and shape. By incorporating the true LoG, Gaussian derivative filters provide a more improved understanding of the spatial structure of the image across multiple scales, which is critical in a medical context where small, complex features can indicate the presence or extent of a tumor.

Another important step is adaptive scale selection, in adaptive scale selection, the algorithm dynamically chooses the most appropriate scale for each keypoint based on the local image content. Instead of using a fixed number of scales, as in traditional SIFT, adaptive scale selection continuously adjusts based on the local gradients, automatically finding the optimal scale at which each feature is most prominent. This allows for better stability in feature detection. In MR images, tumors often appear with unique textures or slightly varying intensities, depending on their type and stage. Adaptive scaling ensures that the algorithm is sensitive to these variations, capturing each feature at its optimal level of detail.

To improve the localization accuracy of tumor features in MR images, subpixel refinement techniques are applied. Traditional SIFT methods detect keypoints at integer pixel coordinates, but subpixel refinement improves this precision by locating keypoints within fractions of a pixel. This approach involves estimating the exact location of the keypoint by analyzing the local intensity values surrounding the keypoint and fitting them to a continuous function. By doing so, the algorithm adjusts the keypoint position based on the interpolated peak within the detected region. This modification allows the detection to be highly accurate, capturing even slight variations in tumor boundaries.

Multiscale orientation assignment is then introduced to handle rotation variations commonly seen in MRI data due to slight head tilts during scanning. In the traditional SIFT algorithm, a single dominant orientation is assigned per keypoint, derived from the direction of the local gradient. However, by using multiscale orientation assignment, the algorithm can assign multiple orientations to each keypoint, accounting for variations in orientation that may occur at different scales. This is achieved by computing orientation histograms at multiple scales and assigning several

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dominant orientations per keypoint. This makes the algorithm rotation invariant, meaning it can detect tumor features consistently regardless of how the patient's head is positioned during the scan.

For more detailed feature capture, extended descriptors with multi-channel histograms are used. Traditional SIFT descriptors capture gradient information in a single intensity channel, which may not be sufficient for representing the complex textures and intensity variations of brain tumors. By using multi-channel histograms, the algorithm incorporates gradient information from multiple intensity ranges. This approach creates a richer feature vector by recording intensity gradients across these channels, allowing for more improved descriptions of the texture and structural complexity within the tumor region. Altogether, these enhancements subpixel refinement, multiscale orientation assignment and multi-channel histogram descriptors work synergistically to create a more precise and rotation-invariant feature extraction process.

To further enhance the feature extraction process for brain tumor detection in MR images, spatial pyramid pooling and RANSAC-based matching play crucial roles in capturing contextual information and ensuring robust matching of keypoints. Spatial pyramid pooling (SPP) enhances the SIFT descriptor by capturing hierarchical spatial information around each keypoint, which is essential for distinguishing tumor regions from normal brain tissue. In traditional feature extraction, keypoint descriptors are often computed only at the local region of each keypoint, limiting the algorithm's awareness of the broader spatial context. SPP overcomes this limitation by dividing the region around each keypoint into increasingly finer sub-regions (pyramidal divisions).

For instance, in a three-level pyramid structure, the area around a keypoint may be divided into a single cell (the whole region) at the coarsest level, four quadrants at the second level, and sixteen sub-regions at the finest level. Within each sub-region, a histogram of gradients is calculated, summarizing gradient orientations in that specific area. These histograms from each level are then concatenated to form an extended feature vector that not only represents local intensity gradients but also encodes spatial relationships across different scales. By doing this, spatial pyramid pooling creates a multi-resolution representation, capturing both fine-grained details (small texture variations within the tumor) and broader contextual patterns (shape or layout of the tumor relative to surrounding tissue). This hierarchical circumstance enables the algorithm to differentiate tumors from surrounding healthy tissue more effectively, as the broader structure of the tumor becomes part of the descriptor.

When comparing features across different images (matching keypoints between pre- and post-contrast MR scans or across different views), outliers in feature matching can introduce errors. RANSAC-based matching is a robust technique for filtering out these mismatches, ensuring that only the most reliable correspondences are used. In brain tumor detection, RANSAC enhances the robustness of matching by iteratively estimating a transformation model based on randomly selected subsets of keypoint pairs.

RANSAC selects a small random sample of keypoint pairs from two images, computes the transformation (translation or rotation) that aligns them, and then evaluates how well this transformation applies to all other keypoint pairs. Keypoint pairs that fit this transformation are classified as inliers, while those that do not fit are

considered outliers and excluded from the final matching. This iteration continues multiple times, selecting new random subsets in each iteration, until RANSAC identifies the transformation with the highest inlier count. This means that RANSAC effectively discards false matches, such as those caused by noise, slight anatomical variations or non-tumor features that may have similar textures. Only consistent matches those that align according to the chosen model are retained. This is particularly beneficial in brain MR images, where slight shifts or rotations can occur between scans, as RANSAC ensures that only genuine tumor-related features are matched across images.

Spatial pyramid pooling adds an improved layer of contextual awareness to the extracted features, which is essential for discerning tumor boundaries and shapes. Meanwhile, RANSAC-based matching ensures that only the most stable and consistent features are carried over during matching, which is invaluable when analyzing tumors and comparing different scans.

Together, these enhancements make the feature extraction process to address the complexities of brain tumor imaging, providing a more accurate, contextually aware and reliable methodology for identifying tumor features across MR images.

3.1 SIFT Algorithm:

i. Scale-Space Construction

- To detect keypoints across multiple scales, SIFT constructs a scale-space representation of the image by convolving it with Gaussians at increasing scales.
- Given an image $I(x, y)$, its scale space $L(x, y, \sigma)$ at scale σ is:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y) \quad (1)$$

$$\text{where, } G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (2)$$

is the Gaussian function. The parameter σ controls the scale.

ii. Difference of Gaussians (DoG)

Keypoints are detected by identifying extrema in the Difference of Gaussians (DoG) images, which approximate the Laplacian of Gaussian.

- The DoG is defined as

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma) \quad (3)$$

where k is a constant factor.

- The DoG is computed at different scales, and keypoints are found by locating local maxima and minima across these scales and neighboring pixels.

iii. Gradient Computation for Orientation Assignment

- Each keypoint is assigned an orientation to achieve rotation invariance.
- For each pixel around the keypoint, gradients are computed using:

$$G_{x(x,y)} = \frac{\partial L(x,y)}{\partial x}, G_{y(x,y)} = \frac{\partial L(x,y)}{\partial y} \quad (4)$$

- The magnitude M and angle θ of the gradient are:

$$M(x, y) = \sqrt{G_{x(x,y)}^2 + G_{y(x,y)}^2}, \quad \theta(x, y) = \arctan\left(\frac{G_{y(x,y)}}{G_{x(x,y)}}\right) \quad (5)$$

- An orientation histogram is created from these angles, weighted by magnitudes, around each keypoint. The peak of the histogram gives the dominant orientation.

iv. Region of Interest (ROI) Masking

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- To focus keypoint detection on a specific area, a mask $M_{ROI}(x,y)$ is created with value 1 within the ROI and 0 elsewhere:

$$M_{ROI(x,y)} = \begin{cases} 1 & \text{if } x_{start} \leq x \leq x_{end} \text{ and } y_{start} \leq y \leq y_{end} \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

- Only keypoints within this mask are processed, making detection spatially selective.

v. Descriptor Calculation

- For each keypoint, a descriptor vector is created by dividing a local region around the keypoint into sub-regions and computing orientation histograms for each.
- The descriptor is a vector of gradient histograms, typically of size $4 \times 4 \times 8 = 128$:

$$\text{Descriptor} = [H_{1,1}, H_{1,2}, \dots, H_{4,4}] \quad (7)$$

where $H_{i,j}$ are the histogram values for each sub-region, capturing both spatial and gradient information.

vi. Keypoint Matching (Optional)

- When comparing two images, keypoints are matched by calculating the Euclidean distance between descriptors:

$$d_{ij} = \sqrt{\sum_{k=1}^{128} (f_{ik} - f_{jk})^2} \quad (8)$$

where d_{ij} is the distance between keypoint i in one image and keypoint j in another, ensuring the best matching features.

3.2 Modified SIFT Algorithm:

i. Improved Scale-Space Representation Using Gaussian Derivative Filters

- Instead of the DoG approximation, use Gaussian derivatives at different scales, which provide a more accurate representation of scale-space and improve localization.
- Construct the scale-space $L(x,y,\sigma)$ using Gaussian derivatives:

$$L(x,y,\sigma) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (G(x,y,\sigma) * I(x,y)) \quad (9)$$

- This change ensures that the algorithm directly computes the LoG, leading to more precise keypoint detection, especially in noisy images.

ii. Adaptive Scale Selection for Keypoint Stability

- Introduce an adaptive scale selection mechanism that adjusts k based on image content. This approach uses local statistics to choose an optimal scale, ensuring that keypoints are detected at the most relevant scales.
- For each region, compute a local scale σ based on image content:

$$\sigma = \sigma_{base} \times (1 + \alpha \times \text{entropy}(I(x,y))) \quad (10)$$

where σ_{base} is the initial scale and α is a scaling factor based on the entropy or variance of $I(x,y)$.

iii. Enhanced Keypoint Localization Using Subpixel Refinement

- Apply subpixel refinement by fitting a quadratic function to the DoG response around each candidate keypoint and refining the location based on the maximum response point.
- For a keypoint candidate at (x_0, y_0, σ) , use Taylor expansion to estimate the true location:

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$$\hat{x} = x_0 - \frac{\frac{\partial D}{\partial x}}{\frac{\partial^2 D}{\partial x^2}} \quad (11)$$

where, $D(x,y,\sigma)$ is the DoG response. This refinement improves accuracy in keypoint localization, especially in low-texture regions.

iv. Multiscale Orientation Assignment for Rotation Robustness

- Use a multiscale orientation assignment that computes gradients at multiple scales around each keypoint, allowing for more robust orientation estimation.
- Compute the gradient magnitudes M and angles θ at each scale and combine them using weighted averaging:

$$M_{\text{multi}(x,y)} = \sum_{s=1}^S w(s) \times M_{s(x,y)} \quad (12)$$

where, $w(s)$ is a scale-dependent weight.

v. Extended Feature Descriptors with Multi-Channel Histograms

- Extend the descriptor to use multi-channel histograms to capture more distinctive patterns.
- Instead of a 4×4 grid with 8 orientation bins, add color formation:

$$\text{Descriptor} = [H_{1,1}^R, H_{1,1}^G, H_{1,1}^B, H_{1,2}^R, \dots, H_{6,6}^B] \quad (13)$$

where $H_{i,j}^C$ is the histogram for color channel C (Red, Green, Blue) in sub-region (i,j) . This modification significantly improves feature distinctiveness in color images.

vi. Spatial Pyramid Pooling for Contextual Information

- Use spatial pyramid pooling (SPP) to capture features at multiple levels, increasing robustness to spatial transformations.
- Divide the image into increasingly fine grids and compute descriptors for each level:

$$\text{SPP - Descriptor} = [\text{Descriptor}_{1 \times 1}, \text{Descriptor}_{2 \times 2}, \text{Descriptor}_{4 \times 4}] \quad (14)$$

This technique captures both local and global information, which can be useful in scenes with complex structures.

vii. Improved Keypoint Matching Using RANSAC and Geometric Verification

- Use RANSAC (Random Sample Consensus) for robust matching by estimating the geometric transformation between images.
- Match keypoints using Euclidean distance, then apply RANSAC to find an optimal homography matrix H that aligns matching points.

$$d_{ij} = \sqrt{\sum_{k=1}^{128} (f_{ik} - f_{jk})^2} \quad (15)$$

where only matches consistent with H are retained, filtering out false matches and improving accuracy.

These improvements provide a more accurate and robust feature extraction framework than standard SIFT, enhancing its performance in challenging environments, such as scenes with high noise, complex patterns or variations in lighting and rotation.

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4 Result and Discussions

This section presents the results obtained using the proposed feature extraction method for brain tumor MR images.

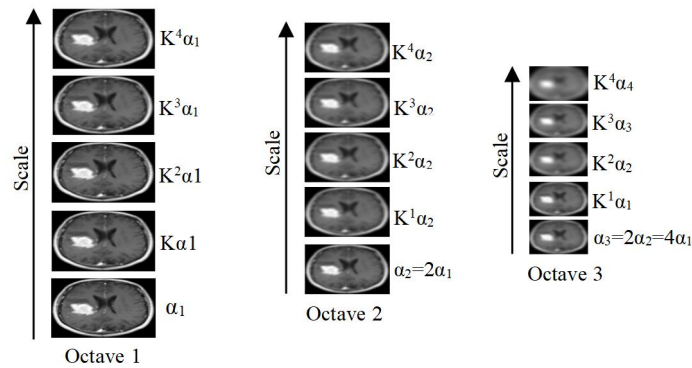


Fig. 2: Progressive blurring of an image across multiple scales and octaves.

Fig. 2 illustrates the progressive blurring of an image across various scales and octaves. Each octave consists of a series of images that are down sampled to lower resolutions. Within each octave, the images are blurred by increasing degrees. In the leftmost column, Octave 1 begins with the original image at the bottom, becoming progressively more blurred as we move upward. The middle column, Octave 2, shows a down sampled version of the image, also blurred to varying extents. Finally, the rightmost column represents Octave 3, where the image is further reduced in size and blurred across different levels.

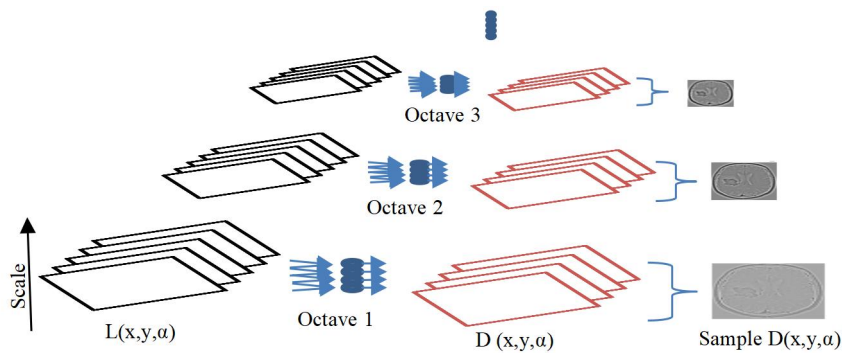


Fig. 3: Generation of Difference of Gaussians (DoG) images through the subtraction of Gaussian-blurred images at different scales.

After the images are blurred, the algorithm generates Difference of Gaussians (DoG) images by subtracting one blurred version from another at different scales as shown in Fig. 3. This process emphasizes areas with notable intensity changes, which are potential keypoints. The DoG highlights regions of significant pixel intensity variation, aiding in keypoint detection.

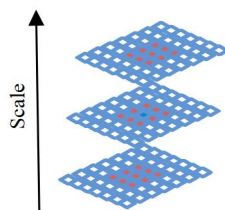


Fig. 4: Keypoint detection by identifying local extrema in Gaussian derivatives images across multiple scales.

The algorithm locates keypoints by identifying local extrema (either bright or dark spots) in the Gaussian derivatives images across both scale and spatial dimensions. As shown in the Fig 4, each pixel in a Gaussian derivatives image is compared to:

- 8 neighboring pixels within the same scale (the same blurred image),
- 9 pixels in the scale above (the preceding blurred image),
- 9 pixels in the scale below (the following blurred image).

In the Fig 5, 5(a) displays the initial keypoints detected across various scales using adaptive scale selection. In 5(b), keypoints with low contrast have been removed, leaving only the stronger, more stable points. Finally, 5(c) shows the result after further refining by eliminating keypoints located on edges, resulting in the final set of well-localized keypoints.

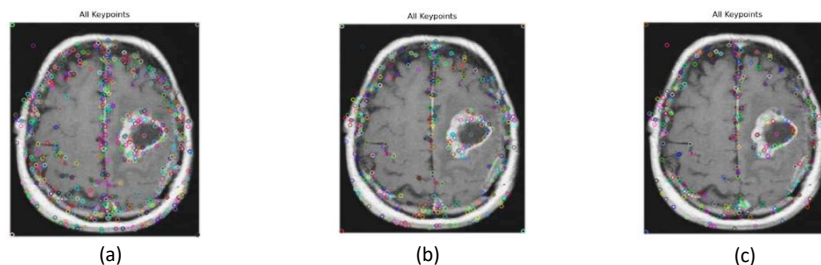


Fig. 5: Keypoint refinement process in Modified SIFT.

Once the scale-space is created and initial keypoints (local maxima or minima in the DoG images) are identified, the keypoint locations need refinement to ensure accuracy. To pinpoint each keypoint more precisely, a mathematical approach called Taylor series expansion is applied. This technique allows for a closer approximation of the exact position of each keypoint, similar to fine-tuning a camera focus for a clearer image. For SIFT, it is used to estimate the Gaussian derivatives function near a keypoint candidate to better define its exact position and scale. Some initial keypoints might lie in areas with insufficient contrast or too little brightness variation, making them susceptible to noise. This ensures that only keypoints with both accurate positioning and sufficient contrast are retained.

Furthermore, while Gaussian derivatives detects strong responses along edges, these points tend to be unreliable. To eliminate such points, the algorithm examines the curvature of the image in different directions around each keypoint. For instance, consider a ridge in a mountain range: the shape along the ridge direction changes very little (low curvature), while across the ridge (up or down), the shape varies significantly (high curvature). If a keypoint shows strong curvature in one direction but not the other, it is rejected as it lacks clear localization. After removing low-contrast and edge-responsive keypoints, the remaining points are well-localized and stable, making them ideal for further processing.

The identified keypoints are regarded as stable, meaning they remain relatively unchanged even with slight modifications to the image. To achieve rotation invariance, each keypoint is assigned a multiscale orientation. This orientation allows

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the algorithm to recognize the keypoint's position even if the image is rotated, ensuring that it can consistently locate the keypoint in the same region as show in Fig 6.



Fig. 6: Orientation assignment for keypoints to ensure rotation invariance using multiscale orientation.

To determine the orientation for each keypoint, a small area surrounding it is analyzed according to its scale, focusing on the gradients and magnitudes of the image. An orientation histogram with 36 bins is generated to represent all possible angles (360 degrees). The peak in this histogram indicates the primary direction of the keypoint. Additionally, any other peaks that reach at least 80% of the highest peak are considered, allowing for the assignment of supplementary orientations. This means that keypoints at the same location and scale can possess multiple orientations, which enhances the robustness of matching keypoints across different images.

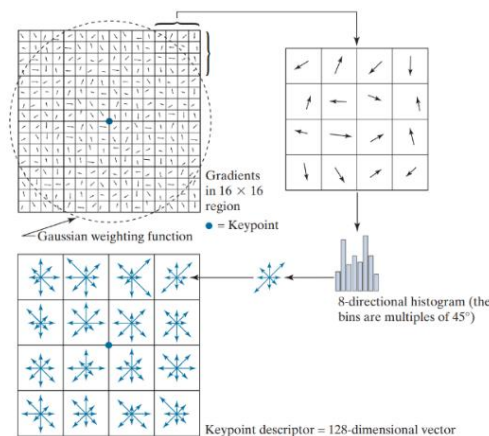


Fig. 7: Computing a keypoint descriptor (Source: Digital Image Processing 4e. Gonzalez and Woods)

Once keypoints have been identified and assigned orientations, the next step involves generating a descriptor for each keypoint as shown in Fig 7. This descriptor serves as a compact representation of the local image information surrounding the keypoint.

To create a keypoint descriptor, a 16x16 pixel region around each keypoint is examined, which is then divided into 16 smaller 4x4 pixel blocks. An 8-bin histogram is constructed for each of these small blocks to capture the distribution of feature orientations. This process results in a total of 128 values that comprise the keypoint descriptor. The descriptor is designed to be invariant to scale and rotation, while also being robust against variations in lighting and viewpoint.

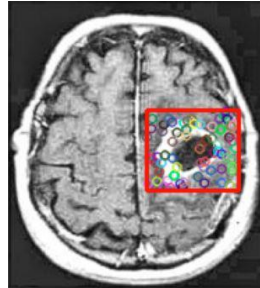


Fig. 8: Detection of Tumor Using the Proposed Feature Extraction Method.

Fig. 8 illustrates the results of tumor detection using the feature extraction method proposed in our study. Highlighted areas indicate the detected tumors, showcasing the effectiveness of the feature extraction approach in improving diagnostic precision.

Table 1. Comparison of Quality Measures for SIFT and MSIFT

Quality measures	SIFT	MSIFT
Precision	0.713	0.770
Recall	0.820	0.886
F1 Score	0.763	0.824
Specificity	0.735	0.796
Mean Squared Error (MSE)	0.226	0.209
Root Mean Squared Error (RMSE)	0.476	0.438
Structural Similarity Index Measure (SSIM)	0.414	0.448
Area Under Curve (AUC) for ROC	0.777	0.840
Dice Coefficient (DC)	0.763	0.824
Jaccard Index	0.616	0.666
Peak Signal-to-Noise Ratio (PSNR)	12.398	13.391
Hausdorff Distance	1752.848	1612.620

Table [1] presents a side-by-side comparison of various quality measures for the SIFT and MSIFT methods. Metrics such as Precision, Recall, F1 Score and Specificity show improvements in MSIFT over SIFT. MSE and RMSE are lower in MSIFT, indicating better performance, while similarity measures SSIM, AUC, Dice Coefficient and Jaccard Index are also higher for MSIFT. Additionally, MSIFT demonstrates improvements in PSNR and a lower Hausdorff Distance, suggesting enhanced accuracy and consistency in MSIFT.

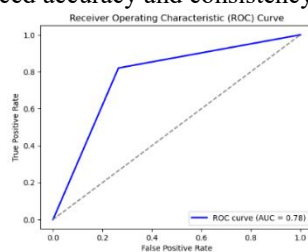


Fig. 9: ROC curve for SIFT

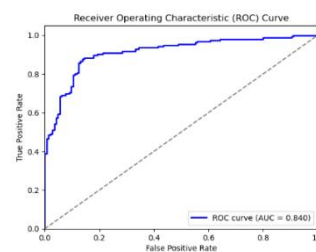


Fig. 10: ROC curve for MSIFT

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Fig.9 illustrates the performance of the SIFT, the Area Under Curve (AUC) value of 0.777 indicates moderate discriminative ability for SIFT in distinguishing between classes. Fig.10 shows the performance of the MSIFT method, the AUC value of 0.840 demonstrates an improvement over SIFT, indicating that MSIFT has better class discrimination capabilities.

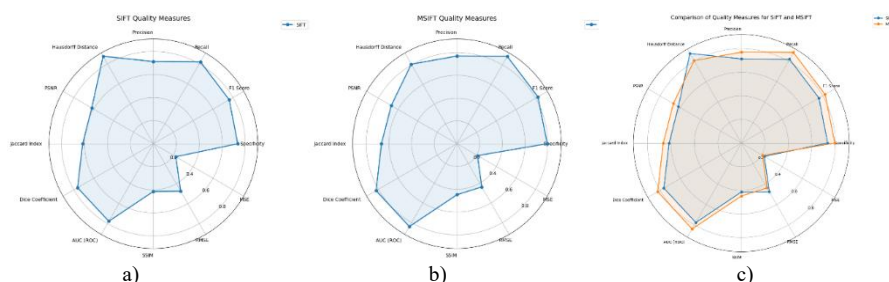


Fig. 11: Comparison of Quality Measures for SIFT and MSIFT

In Fig.11, 11(a) presents a radar graph illustrating the quality measures for SIFT, 11(b) shows a radar graph for the quality measures of MSIFT and 11(c) provides a radar graph that compares the quality measures of both SIFT and MSIFT.

5 Conclusion

In this paper, we introduced a modified version of the Scale-Invariant Feature Transform (SIFT) algorithm, termed MSIFT, which significantly enhances feature extraction for medical image analysis, specifically for brain tumor detection in MR images. By incorporating adaptive scale selection, subpixel refinement, multiscale orientation assignment and extended feature descriptors, MSIFT improves upon the original SIFT algorithm in terms of both accuracy and robustness. Quantitative results demonstrate that MSIFT outperforms SIFT across several key metrics, including Precision, Recall, F1-Score and Specificity, with notable improvements in image quality metrics like SSIM, AUC, Dice Coefficient and Jaccard Index. Error-based measures such as MSE, RMSE and Hausdorff Distance also indicate superior performance of MSIF. These results highlight the proposed method's ability to provide more accurate and reliable feature extraction.

Future work after MSIFT feature extraction on MR images includes optimizing feature selection, combining with other descriptors, integrating with deep learning, validating on large datasets, extending to 3D MRI.

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